

**LAPORAN PRAKTIKUM
PEMROGRAMAN MOBILE
MODUL 4**



VIEWMODEL AND DEBUGGING

Oleh:

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**PROGRAM STUDI TEKNOLOGI INFORMASI
FAKULTAS TEKNIK
UNIVERSITAS LAMBUNG MANGKURAT
MEI 2025**

LEMBAR PENGESAHAN
LAPORAN PRAKTIKUM PEMROGRAMAN MOBILE
MODUL 4

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DAFTAR ISI

LEMBAR PENGESAHAN.....	2
DAFTAR ISI	3
DAFTAR GAMBAR.....	4
DAFTAR TABEL	5
SOAL 1.....	6
A. Source Code	6
B. Output Program.....	30
C. Pembahasan	31
D. Tautan Git	32
SOAL 2.....	32
A. Pembahasan	32

DAFTAR GAMBAR

Gambar 1 Screenshot Hasil Jawaban Soal 1 XML	30
Gambar 2 Screenshot Hasil Jawaban Soal 1 Jetpack Compose	30

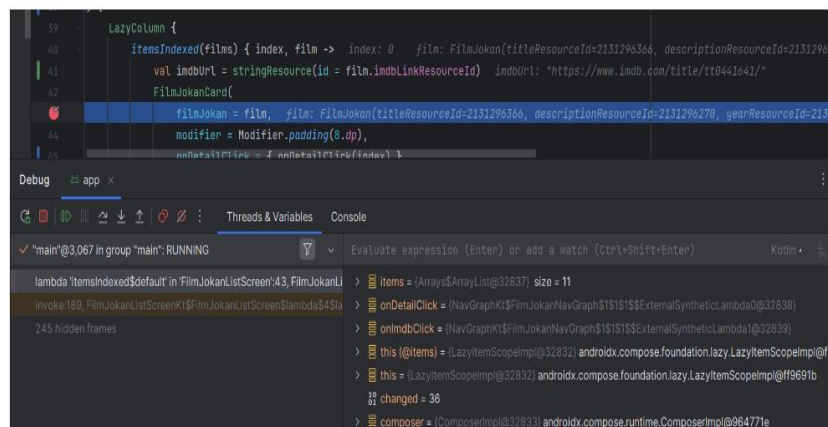
DAFTAR TABEL

Tabel 1 Source Code Jawaban Soal 1.....	12
Tabel 2 Source Code Jawaban Soal 1.....	18
Tabel 3. Source Code Jawaban Soal 1.....	26

Soal Praktikum:

SOAL 1

1. Lanjutkan aplikasi Android berbasis XML dan Jetpack Compose yang sudah dibuat pada Modul 3 dengan menambahkan modifikasi sesuai ketentuan berikut:
 - a. Buatlah sebuah ViewModel untuk menyimpan dan mengelola data dari list item. Data tidak boleh disimpan langsung di dalam Fragment atau Activity.
 - b. Gunakan ViewModelFactory untuk membuat parameter dengan tipe data String di dalam ViewModel
 - c. Gunakan StateFlow untuk mengelola event onClick dan data list item dari ViewModel ke Fragment
 - d. Install dan gunakan library Timber untuk logging event berikut:
 - Log saat data item masuk ke dalam list
 - Log saat tombol Detail dan tombol Explicit Intent ditekan
 - Log data dari list yang dipilih ketika berpindah ke halaman Detail
 - Gunakan tool Debugger di Android Studio untuk melakukan debugging pada aplikasi.
 - e. Cari setidaknya satu breakpoint yang relevan dengan aplikasi. Lalu, gunakan fitur Step Into, Step Over, dan Step Out. Setelah itu, jelaskan fungsi Debugger, cara menggunakan Debugger, serta fitur Step Into, Step Over, dan Step Out
2. Jelaskan Application class dalam arsitektur aplikasi Android dan fungsinya
Aplikasi harus dapat mempertahankan fitur-fitur yang sudah dibuat pada modul sebelumnya. Berikut adalah contoh debugging dalam Android Studio.



Gambar 1 Contoh Penggunaan Debugger

A. Source Code

Kode Jetpack Compose

GameDetailScreen.kt

```
1 package com.example.modul4compose.ui.screen
```

```

2
3 import android.content.Intent
4 import android.net.Uri
5 import androidx.compose.foundation.background
6 import androidx.compose.foundation.layout.*
7 import androidx.compose.foundation.rememberScrollState
8 import androidx.compose.foundation.verticalScroll
9 import androidx.compose.material.icons.Icons
10 import androidx.compose.material.icons.filled.ArrowBack
11 import androidx.compose.material3.*
12 import androidx.compose.runtime.Composable
13 import androidx.compose.runtime.collectAsState
14 import androidx.compose.runtime.getValue
15 import androidx.compose.ui.Alignment
16 import androidx.compose.ui.Modifier
17 import androidx.compose.ui.layout.ContentScale
18 import androidx.compose.ui.platform.LocalContext
19 import androidx.compose.ui.text.style.TextAlign
20 import androidx.compose.ui.unit.dp
21 import androidx.navigation.NavController
22 import coil.compose.AsyncImage
23 import com.example.modul4compose.viewmodel.GamesViewModel
24 import androidx.compose.ui.graphics.Brush
25 import androidx.compose.ui.graphics.Color
26 import timber.log.Timber
27
28 @OptIn(ExperimentalMaterial3Api::class)
29 @Composable
30 fun GameDetailScreen(
31     gameId: Int,
32     navController: NavController,
33     viewModel: GamesViewModel,
34     modifier: Modifier = Modifier
35 ) {
36     val selectedGame by viewModel.selectedGame.collectAsState()
37     val context = LocalContext.current
38
39
40     val game = selectedGame ?: viewModel.getGameById(gameId)
41
42     game?.let {
43         Timber.d("Viewing game detail: ${it.title}")
44         Scaffold(
45             topBar = {
46                 TopAppBar(
47                     title = { Text(text = game.title) },
48                     navigationIcon = {

```

```

49         IconButton(onClick = {
navController.navigateUp() }) {
50             Icon(Icons.Default.ArrowBack,
contentDescription = "Back")
51         }
52     },
53     colors = TopAppBarDefaults.topAppBarColors(
54         containerColor =
MaterialTheme.colorScheme.primaryContainer,
55         titleContentColor =
MaterialTheme.colorScheme.onPrimaryContainer
56     )
57 )
58 }
59 ) { paddingValues ->
60     Column(
61         modifier = modifier
62             .fillMaxSize()
63             .padding(paddingValues)
64             .verticalScroll(rememberScrollState())
65     ) {
66
67         Box(
68             modifier = Modifier
69                 .fillMaxWidth()
70                 .height(250.dp)
71         ) {
72             AsyncImage(
73                 model = game.imageUrl,
74                 contentDescription = game.title,
75                 modifier = Modifier.fillMaxSize(),
76                 contentScale = ContentScale.Crop
77             )
78
79             Box(
80                 modifier = Modifier
81                     .fillMaxSize()
82                     .background(
83                         brush = Brush.verticalGradient(
84                             colors = listOf(
85                                 Color.Transparent,
86                                 Color.Black.copy(alpha =
0.7f)
87                             )
88                         )
89                     )
90             )

```



```

91
92         Column(
93             modifier = Modifier
94                 .align(Alignment.BottomStart)
95                 .padding(16.dp)
96         ) {
97             Text(
98                 text = game.title,
99                 style =
MaterialTheme.typography.headlineMedium,
100                 color = Color.White
101             )
102             Text(
103                 text = game.playersCount,
104                 style =
MaterialTheme.typography.titleMedium,
105                 color = Color.White.copy(alpha = 0.8f)
106             )
107         }
108
109
110         Column(
111             modifier = Modifier
112                 .padding(16.dp)
113                 .fillMaxWidth()
114         ) {
115
116             ElevatedCard(
117                 modifier = Modifier.fillMaxWidth(),
118                 colors = CardDefaults.elevatedCardColors(
119                     containerColor =
MaterialTheme.colorScheme.surfaceVariant
120                 )
121             ) {
122                 Row(
123                     modifier = Modifier
124                         .fillMaxWidth()
125                         .padding(16.dp),
126                     horizontalArrangement =
Arrangement.SpaceEvenly
127                 ) {
128                     InfoColumn("Genre", game.genre)
129                     VerticalDivider()
130                     InfoColumn("Rating", game.rating)
131                     VerticalDivider()

```

132	InfoColumn("Released",	
	game.releaseDate)	
133	}	
134	}	
135		
136	Spacer(modifier = Modifier.height(16.dp))	
137		
138		
139	Text(	
140	text = "About",	
141	style	=
	MaterialTheme.typography.titleLarge,	
142	color	=
	MaterialTheme.colorScheme.onSurface	
143	)	
144	Spacer(modifier = Modifier.height(8.dp))	
145	Text(	
146	text = game.description,	
147	style	=
	MaterialTheme.typography.bodyLarge,	
148	color	=
	MaterialTheme.colorScheme.onSurfaceVariant	
149	)	
150		
151	Spacer(modifier = Modifier.height(16.dp))	
152		
153		
154	Text(	
155	text = "Developer",	
156	style	=
	MaterialTheme.typography.titleLarge,	
157	color	=
	MaterialTheme.colorScheme.onSurface	
158	)	
159	Spacer(modifier = Modifier.height(8.dp))	
160	Text(	
161	text = game.developer,	
162	style	=
	MaterialTheme.typography.bodyLarge,	
163	color	=
	MaterialTheme.colorScheme.onSurfaceVariant	
164	)	
165		
166	Spacer(modifier = Modifier.height(24.dp))	
167		
168		
169	Button(	

```

170         onClick = {
171             Timber.d("Explicit Intent button
clicked in detail screen for: ${game.title}")
172             val intent =
Intent(Intent.ACTION_VIEW, Uri.parse(game.gameUrl))
173             context.startActivity(intent)
174         },
175         modifier = Modifier.fillMaxWidth(),
176         colors = ButtonDefaults.buttonColors(
177             containerColor =
MaterialTheme.colorScheme.primary
178         )
179     ) {
180         Text(
181             text = "Play Now",
182             style =
MaterialTheme.typography.titleMedium
183         )
184     }
185 }
186 }
187 }
188 } ?: run {
189     Box(
190         modifier = Modifier
191             .fillMaxSize()
192             .padding(16.dp)
193     ) {
194         Text("Game not found")
195     }
196 }
197 }
198
199 @Composable
200 private fun InfoColumn(
201     label: String,
202     value: String,
203     modifier: Modifier = Modifier
204 ) {
205     Column(
206         horizontalAlignment = Alignment.CenterHorizontally,
207         modifier = modifier
208     ) {
209         Text(
210             text = label,
211             style = MaterialTheme.typography.labelMedium,
212             color = MaterialTheme.colorScheme.onSurfaceVariant

```

213	)
214	Spacer(modifier = Modifier.height(4.dp))
215	Text(
216	text = value,
217	style = MaterialTheme.typography.bodyMedium,
218	color = MaterialTheme.colorScheme.onSurface,
219	textAlign = TextAlign.Center
220	)
221	}
222	}
223	
224	@Composable
225	private fun VerticalDivider(
226	modifier: Modifier = Modifier
227	) {
228	Box(
229	modifier = modifier
230	.width(1.dp)
231	.height(32.dp)
232	
233	.background(MaterialTheme.colorScheme.onSurfaceVariant.copy(alpha
	= 0.2f))
234	)
235	}

Tabel 1 Source Code Jawaban Soal 1

GameListScreen.kt

1	package com.example.modul4compose.ui.screen
2	
3	import android.content.Intent
4	import android.net.Uri
5	import androidx.compose.foundation.layout.*
6	import androidx.compose.foundation.lazy.LazyColumn
7	import androidx.compose.foundation.lazy.items
8	import androidx.compose.foundation.shape.RoundedCornerShape
9	import androidx.compose.material3.*
10	import androidx.compose.runtime.Composable
11	import androidx.compose.runtime.collectAsState
12	import androidx.compose.runtime.getValue
13	import androidx.compose.ui.Alignment
14	import androidx.compose.ui.Modifier
15	import androidx.compose.ui.draw.clip
16	import androidx.compose.ui.layout.ContentScale
17	import androidx.compose.ui.platform.LocalContext
18	import androidx.compose.ui.text.style.TextOverflow
19	import androidx.compose.ui.unit.dp
20	import androidx.lifecycle.viewmodel.compose.viewModel

```

21 import androidx.navigation.NavController
22 import coil.compose.AsyncImage
23 import com.example.modul4compose.data.RobloxGamesData
24 import com.example.modul4compose.navigation.Screen
25 import com.example.modul4compose.viewmodel.GameViewModel
26 import
    com.example.modul4compose.viewmodel.GameViewModelFactory
27 import timber.log.Timber
28
29 @Composable
30 fun GamesListScreen(
31     navController: NavController,
32     viewModel: GameViewModel,
33     modifier: Modifier = Modifier
34 ) {
35     val games by viewModel.games.collectAsState()
36
37     LazyColumn(
38         modifier = modifier.fillMaxSize(),
39         contentPadding = PaddingValues(16.dp),
40         verticalArrangement = Arrangement.spacedBy(16.dp)
41     ) {
42         items(games) { game ->
43             val context = LocalContext.current
44             GameCard(
45                 game = game,
46                 onPlayClick = {
47                     Timber.d("Explicit Intent button clicked
for: ${game.title}")
48                     val intent = Intent(Intent.ACTION_VIEW,
Uri.parse(game.gameUrl))
49                     context.startActivity(intent)
50                 },
51                 onDetailsClick = {
52                     Timber.d("Detail button clicked for:
${game.title}")
53                     viewModel.selectGame(game)
54
55                 navController.navigate(Screen.GameDetail.createRoute(game.id))
56             }
57         }
58     }
59 }
60
61 @OptIn(ExperimentalMaterial3Api::class)
62 @Composable

```

```

63 private fun GameCard(
64     game: com.example.modul4compose.data.RobloxGame,
65     onPlayClick: () -> Unit,
66     onDetailsClick: () -> Unit,
67     modifier: Modifier = Modifier
68 ) {
69     Card(
70         modifier = modifier.fillMaxWidth(),
71         shape = RoundedCornerShape(16.dp)
72     ) {
73         Row(
74             modifier = Modifier
75                 .padding(16.dp)
76                 .fillMaxWidth(),
77             verticalAlignment = Alignment.CenterVertically
78         ) {
79             AsyncImage(
80                 model = game.imageUrl,
81                 contentDescription = game.title,
82                 modifier = Modifier
83                     .size(width = 120.dp, height = 120.dp)
84                     .clip(RoundedCornerShape(12.dp)),
85                 contentScale = ContentScale.Crop
86             )
87
88             Spacer(modifier = Modifier.width(16.dp))
89
90             Column(
91                 modifier = Modifier.weight(1f)
92             ) {
93                 Text(
94                     text = game.title,
95                     style = MaterialTheme.typography.titleLarge
96                 )
97
98                 Spacer(modifier = Modifier.height(4.dp))
99
100                Text(
101                    text = game.playersCount,
102                    style = MaterialTheme.typography.titleMedium,
103                    color = MaterialTheme.colorScheme.primary
104                )
105
106                Spacer(modifier = Modifier.height(4.dp))
107

```

108	Text(	
109	text = game.genre,	
110	style	=
	MaterialTheme.typography.bodyMedium,	
111	color	=
	MaterialTheme.colorScheme.onSurfaceVariant	
112	)	
113		
114	Text(	
115	text = game.rating,	
116	style	=
	MaterialTheme.typography.bodyMedium,	
117	color	=
	MaterialTheme.colorScheme.onSurfaceVariant	
118	)	
119		
120	Spacer(modifier = Modifier.height(8.dp))	
121		
122	Row(	
123	modifier = Modifier.fillMaxWidth(),	
124	horizontalArrangement	=
	Arrangement.spacedBy(8.dp)	
125	) {	
126	Button(	
127	onClick = onPlayClick,	
128	modifier = Modifier.weight(1f),	
129	colors = ButtonDefaults.buttonColors(	
130	containerColor	=
	MaterialTheme.colorScheme.primary,	
131	contentColor	=
	MaterialTheme.colorScheme.onPrimary	
132	)	
133	) {	
134	Text(	
135	text = "Play Now",	
136	maxLines = 1,	
137	overflow = TextOverflow.Ellipsis	
138	)	
139	}	
140		
141	Button(	
142	onClick = onDetailsClick,	
143	modifier = Modifier.weight(1f),	
144	colors = ButtonDefaults.buttonColors(	
145	containerColor	=
	MaterialTheme.colorScheme.secondary,	

```

146                                     contentColor
147 MaterialTheme.colorScheme.onSecondary
148                                     )
149                                     ) {
150                                     Text(
151                                         text = "View Details",
152                                         maxLines = 1,
153                                         overflow = TextOverflow.Ellipsis
154                                     )
155                                     }
156                                 }
157                            }
158                    }
159            }
160    }

```

Tabel 2 Source Code Jawaban Soal 1

RobloxGame.kt

```

1 package com.example.modul4compose.data
2
3 data class RobloxGame(
4     val id: Int,
5     val title: String,
6     val description: String,
7     val imageUrl: String,
8     val playersCount: String,
9     val gameId: String,
10    val genre: String,
11    val rating: String,
12    val releaseDate: String,
13    val developer: String
14 )
15
16 object RobloxGamesData {
17     val games = listOf(
18         RobloxGame(
19             1,
20             "Adopt Me!",
21             "A role-playing game where players can adopt and
raise pets, decorate homes, and interact with other players.
Features include pet trading, house customization, and mini-
games.",
22             "https://tr.rbxcdn.com/180DAY-
23a22e242c44352464dbca03bbcb189a/768/432/Image/Webp/noFilter",
23             "500K+ Active Players",
24             "https://www.roblox.com/games/920587237",
25             "Role-Playing",

```



```

26         "4.5/5",
27         "July 2017",
28         "DreamCraft"
29     ),
30     RobloxGame(
31         2,
32         "Brookhaven RP",
33         "A roleplaying game where players can live, work,
and socialize in a virtual city. Features include house
ownership, vehicle system, and various jobs to choose from.",
34         "https://tr.rbxcdn.com/180DAY-
d743ef79fbc0a90f9aea0bdf740cd00c/768/432/Image/Webp/noFilter",
35         "450K+ Active Players",
36         "https://www.roblox.com/games/4924922222",
37         "Role-Playing",
38         "4.3/5",
39         "April 2020",
40         "Wolfpaq"
41     ),
42     RobloxGame(
43         3,
44         "Blox Fruits",
45         "An action-adventure game inspired by One Piece
where players can train, fight, and collect Devil Fruits.
Features include PvP combat, fruit hunting, and character
progression.",
46         "https://tr.rbxcdn.com/180DAY-
37cc49a6bdb03b5a4394db84ff264fe5/768/432/Image/Webp/noFilter",
47         "400K+ Active Players",
48         "https://www.roblox.com/games/2753915549",
49         "Action-Adventure",
50         "4.7/5",
51         "January 2019",
52         "Gamer Robot Inc"
53     ),
54     RobloxGame(
55         4,
56         "Pet Simulator X",
57         "Collect pets, hatch eggs, and explore various
worlds while becoming the ultimate pet master. Features include
trading system, rare pets collection, and world exploration.",
58         "https://tr.rbxcdn.com/180DAY-
384dc534a8bd24e4c6217056c1d4ad4f/768/432/Image/Webp/noFilter",
59         "300K+ Active Players",
60         "https://www.roblox.com/games/6284583030",
61         "Simulation",
62         "4.4/5",

```

```

63         "July 2021",
64         "BIG Games"
65     ),
66     RobloxGame(
67         5,
68         "Murder Mystery 2",
69         "A thrilling game where players are assigned roles
of innocent, sheriff, or murderer in each round. Features
include weapon collecting, trading system, and competitive
gameplay.",
70         "https://tr.rbxcdn.com/180DAY-
c929c9ba9069190914f60bbfe47b6cb9/768/432/Image/Webp/noFilter",
71         "250K+ Active Players",
72         "https://www.roblox.com/games/142823291",
73         "Action",
74         "4.6/5",
75         "January 2014",
76         "Nikilis"
77     )
78 )
79 }

```

Tabel 3 Source Code Jawaban Soal 1

NavGraph.kt

```

1 package com.example.modul4compose.navigation
2
3 sealed class Screen(val route: String) {
4     object GamesList : Screen("games_list")
5     object GameDetail : Screen("game_detail/{gameId}") {
6         fun createRoute(gameId: Int) =
"game_detail/$gameId"
7     }
8 }
9
10 object NavArguments {
11     const val GAME_ID = "gameId"
12 }

```

Tabel 4 Source Code Jawaban Soal 1

GamesViewModel.kt

```

1 package com.example.modul4compose.viewmodel
2
3 import androidx.lifecycle.ViewModel
4 import kotlinx.coroutines.flow.MutableStateFlow
5 import kotlinx.coroutines.flow.StateFlow
6 import com.example.modul4compose.data.RobloxGame
7 import com.example.modul4compose.data.RobloxGamesData

```

```

8 import timber.log.Timber
9
10 class GamesViewModel(private val username: String) :
    ViewModel() {
11
12     private val _games =
    MutableStateFlow<List<RobloxGame>>(emptyList())
13     val games: StateFlow<List<RobloxGame>> = _games
14
15     private val _selectedGame =
    MutableStateFlow<RobloxGame?>(null)
16     val selectedGame: StateFlow<RobloxGame?> =
    _selectedGame
17
18     init {
19         Timber.i("ViewModel initialized with user:
    $username")
20
21         _games.value = RobloxGamesData.games
22         Timber.i("Loaded ${_games.value.size} games into
    the list")
23     }
24
25     fun selectGame(game: RobloxGame) {
26         _selectedGame.value = game
27         Timber.i("Selected game for detail view:
    ${game.title}")
28     }
29
30     fun gameIdByGame(gameId: Int): RobloxGame? {
31         return _games.value.find { it.id == gameId }
32     }
33 }

```

Tabel 5 Source Code Jawaban Soal 1

GamesViewModelFactory.kt

```

1 package com.example.modul4compose.viewmodel
2
3 import androidx.lifecycle.ViewModel
4 import androidx.lifecycle.ViewModelProvider
5
6 class GamesViewModelFactory(private val username: String)
    : ViewModelProvider.Factory {
7     override fun <T : ViewModel> create(modelClass:
    Class<T>): T {

```

8	<pre> if (modelClass.isAssignableFrom(GamesViewModel::class.java)) { 9 @Suppress("UNCHECKED_CAST") 10 return GamesViewModel(username) as T 11 } 12 throw IllegalArgumentException("Unknown ViewModel class") 13 } 14 } </pre>
---	--

Tabel 6 Source Code Jawaban Soal 1

XML

activity_main.xml

1	<?xml version="1.0" encoding="utf-8"?>
2	<androidx.constraintlayout.widget.ConstraintLayout
3	xmlns:android="http://schemas.android.com/apk/res/android"
4	xmlns:app="http://schemas.android.com/apk/res-auto"
5	android:layout_width="match_parent"
6	android:layout_height="match_parent">
7	
8	<androidx.fragment.app.FragmentContainerView
9	android:id="@+id/nav_host_fragment"
10	android:name="androidx.navigation.fragment.NavHostFragment"
11	android:layout_width="match_parent"
12	android:layout_height="match_parent"
13	app:defaultNavHost="true"
14	app:navGraph="@navigation/nav_graph" />
15	
16	</androidx.constraintlayout.widget.ConstraintLayout>

Tabel 7 Source Code Jawaban Soal 1

fragment_roblox_game_detail.xml

1	<?xml version="1.0" encoding="utf-8"?>
2	<androidx.core.widget.NestedScrollView
3	xmlns:android="http://schemas.android.com/apk/res/an
4	droid"
5	xmlns:app="http://schemas.android.com/apk/res-
6	auto"
7	xmlns:tools="http://schemas.android.com/tools"
8	android:layout_width="match_parent"

```

9         android:layout_height="match_parent"
10        android:fillViewport="true">
11
12
13    <androidx.constraintlayout.widget.ConstraintLayout
14        android:layout_width="match_parent"
15        android:layout_height="wrap_content">
16
17
18        <ImageView
19            android:id="@+id/gameHeaderImage"
20            android:layout_width="match_parent"
21            android:layout_height="200dp"
22            android:scaleType="centerCrop"
23
24            app:layout_constraintTop_toTopOf="parent"
25                tools:src="@drawable/adopt_me"/>
26
27
28        <TextView
29            android:id="@+id/gameTitle"
30            android:layout_width="match_parent"
31            android:layout_height="wrap_content"
32            android:layout_marginHorizontal="16dp"
33            android:layout_marginTop="16dp"
34            android:textAppearance="@style/TextAppearance.MaterialComponents.Headline5"
35            android:textStyle="bold"
36
37            app:layout_constraintTop_toBottomOf="@id/gameHeaderImage"
38            tools:text="Adopt Me!"/>
39
40
41
42    <com.google.android.material.card.MaterialCardView
43        android:id="@+id/infoContainer"
44        android:layout_width="match_parent"
45        android:layout_height="wrap_content"
46        android:layout_margin="16dp"
47        app:cardCornerRadius="8dp"
48        app:cardElevation="4dp"
49
50        app:layout_constraintTop_toBottomOf="@id/gameTitle">
51
52        <LinearLayout

```

```

53         android:layout_width="match_parent"
54         android:layout_height="wrap_content"
55         android:orientation="horizontal"
56         android:padding="16dp"
57         android:weightSum="3">
58
59
60         <LinearLayout
61             android:layout_width="0dp"
62
63         android:layout_height="wrap_content"
64             android:layout_weight="1"
65             android:gravity="center"
66             android:orientation="vertical">
67
68             <TextView
69
70         android:layout_width="wrap_content"
71
72         android:layout_height="wrap_content"
73             android:text="Genre"
74
75         android:textAppearance="@style/TextAppearance.Material
76         alComponents.Caption"/>
77
78             <TextView
79                 android:id="@+id/gameGenre"
80
81         android:layout_width="wrap_content"
82
83         android:layout_height="wrap_content"
84
85         android:layout_marginTop="4dp"
86
87         android:textAppearance="@style/TextAppearance.Material
88         alComponents.Body1"
89             tools:text="Role-Playing"/>
        </LinearLayout>
    </LinearLayout>
    <LinearLayout
        android:layout_width="0dp"
        android:layout_height="wrap_content"
        android:layout_weight="1"
        android:gravity="center"
        android:orientation="vertical">

```

```

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
                android:text="Rating"

            android:textAppearance="@style/TextAppearance.MaterialComponents.Caption"/>

        <TextView
            android:id="@+id/gameRating"

            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="4dp"

            android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"
                tools:text="4.5/5"/>
    </LinearLayout>

    <LinearLayout
        android:layout_width="0dp"

        android:layout_height="wrap_content"
            android:layout_weight="1"
            android:gravity="center"
            android:orientation="vertical">

        <TextView

            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
                android:text="Released"

            android:textAppearance="@style/TextAppearance.MaterialComponents.Caption"/>

        <TextView

            android:id="@+id/gameReleased"

```

```

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:layout_marginTop="4dp"

        android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"
                                tools:text="July 2017"/>
        </LinearLayout>

        </LinearLayout>

</com.google.android.material.card.MaterialCardView>

        <TextView
            android:id="@+id/aboutTitle"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="16dp"
            android:layout_marginTop="24dp"
            android:text="About"

            android:textAppearance="@style/TextAppearance.MaterialComponents.Headline6"

            app:layout_constraintStart_toStartOf="parent"

            app:layout_constraintTop_toBottomOf="@id/infoContainer"/>

        <TextView
            android:id="@+id/gameDescription"
            android:layout_width="match_parent"
            android:layout_height="wrap_content"
            android:layout_marginHorizontal="16dp"
            android:layout_marginTop="8dp"

            android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"

            app:layout_constraintTop_toBottomOf="@id/aboutTitle"
                                tools:text="A role-playing game where
players can adopt

```


and raise pets, decorate homes, and interact with other players.

Features include pet trading, house customization, and mini-games."/>

```
<TextView
    android:id="@+id/developerTitle"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginStart="16dp"
    android:layout_marginTop="24dp"
    android:text="Developer"
```

```
android:textAppearance="@style/TextAppearance.MaterialComponents.Headline6"
```

```
app:layout_constraintStart_toStartOf="parent"
```

```
app:layout_constraintTop_toBottomOf="@id/gameDescription"/>
```

```
<TextView
    android:id="@+id/gameDeveloper"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginHorizontal="16dp"
    android:layout_marginTop="8dp"
```

```
android:textAppearance="@style/TextAppearance.MaterialComponents.Body1"
```

```
app:layout_constraintTop_toBottomOf="@id/developerTitle"
```

```
tools:text="DreamCraft"/>
```

```
<com.google.android.material.button.MaterialButton
    android:id="@+id/playNowButton"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_margin="16dp"
    android:padding="12dp"
    android:text="Play Now"
    android:textSize="16sp"
```

	<pre> app:layout_constraintBottom_toBottomOf="parent" app:layout_constraintTop_toBottomOf="@id/gameDeveloper" app:layout_constraintVertical_bias="1.0"/> </androidx.constraintlayout.widget.ConstraintLayout> </androidx.core.widget.NestedScrollView> </pre>
--	--

Tabel 8. Source Code Jawaban Soal 1

fragment_roblox_game_list.xml

1	<?xml version="1.0" encoding="utf-8"?>
2	<androidx.constraintlayout.widget.ConstraintLayout
3	xmlns:android="http://schemas.android.com/apk/res/android"
4	xmlns:app="http://schemas.android.com/apk/res-auto"
5	android:layout_width="match_parent"
6	android:layout_height="match_parent">
7	
8	<androidx.recyclerview.widget.RecyclerView
9	android:id="@+id/gamesRecyclerView"
10	android:layout_width="match_parent"
11	android:layout_height="match_parent"
12	android:clipToPadding="false"
13	android:padding="8dp"
14	app:layoutManager="androidx.recyclerview.widget.LinearLayoutManager"
15	app:layout_constraintBottom_toBottomOf="parent"
16	app:layout_constraintEnd_toEndOf="parent"
17	app:layout_constraintStart_toStartOf="parent"
18	app:layout_constraintTop_toTopOf="parent" />
19	
20	</androidx.constraintlayout.widget.ConstraintLayout>

Tabel 9 Source Code Jawaban Soal 1

item_roblox_game.xml

1	<?xml version="1.0" encoding="utf-8"?>
2	<com.google.android.material.card.MaterialCardView

```

3  xmlns:android="http://schemas.android.com/apk/res/an
   droid"
4      xmlns:app="http://schemas.android.com/apk/res-
   auto"
5      xmlns:tools="http://schemas.android.com/tools"
6      android:layout_width="match_parent"
7      android:layout_height="wrap_content"
8      android:layout_margin="8dp"
9      app:cardCornerRadius="12dp"
10     app:cardElevation="4dp">
11
12
13     <androidx.constraintlayout.widget.ConstraintLayout
14         android:layout_width="match_parent"
15         android:layout_height="wrap_content"
16         android:padding="8dp">
17
18         <com.google.android.material.imageview.ShapeableImag
19         eView
20             android:id="@+id/gameImage"
21             android:layout_width="120dp"
22             android:layout_height="120dp"
23             android:scaleType="centerCrop"
24
25             app:layout_constraintStart_toStartOf="parent"
26
27             app:layout_constraintTop_toTopOf="parent"
28
29             app:shapeAppearanceOverlay="@style/RoundedCornerImag
30             e"
31
32             tools:src="@drawable/placeholder"/>
33
34         <TextView
35             android:id="@+id/gameTitle"
36             android:layout_width="0dp"
37             android:layout_height="wrap_content"
38             android:layout_marginStart="12dp"
39             android:textSize="18sp"
40             android:textStyle="bold"
41
42             app:layout_constraintEnd_toEndOf="parent"
43
44             app:layout_constraintStart_toEndOf="@id/gameImage"
45
46             app:layout_constraintTop_toTopOf="parent"

```

```

        tools:text="Adopt Me!"/>

        <TextView
            android:id="@+id/gameGenre"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="12dp"
            android:layout_marginTop="4dp"
            android:textSize="14sp"

app:layout_constraintStart_toEndOf="@id/gameImage"

app:layout_constraintTop_toBottomOf="@id/gameTitle"
        tools:text="Role-Playing"/>

        <TextView
            android:id="@+id/gamePlayers"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="12dp"
            android:textColor="#674fa3"
            android:textSize="14sp"

app:layout_constraintStart_toEndOf="@id/gameGenre"

app:layout_constraintTop_toTopOf="@id/gameGenre"
        tools:text="500K+ Players" />

        <TextView
            android:id="@+id/gameRating"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginStart="12dp"
            android:layout_marginTop="4dp"
            android:textSize="14sp"

app:layout_constraintStart_toEndOf="@id/gameImage"

app:layout_constraintTop_toBottomOf="@id/gameGenre"
        tools:text="Rating: 4.5/5"/>

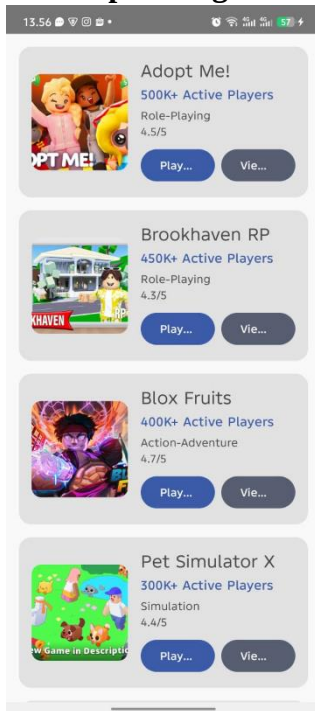
        <Button
            android:id="@+id/playNowButton"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_marginTop="8dp"
            android:backgroundTint="#674fa3"

```

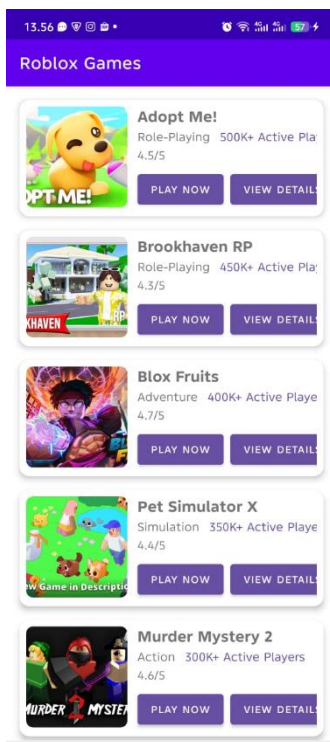
	<pre> android:text="Play Now" android:textSize="12sp" app:layout_constraintStart_toStartOf="@id/gameGenre" app:layout_constraintTop_toBottomOf="@id/gameRating" /> <Button android:id="@+id/viewDetailsButton" android:layout_width="wrap_content" android:layout_height="wrap_content" android:layout_marginStart="8dp" android:backgroundTint="#674fa3" android:text="View Details" android:textSize="12sp" app:layout_constraintStart_toEndOf="@id/playNowButton" n" app:layout_constraintTop_toTopOf="@id/playNowButton" /> </androidx.constraintlayout.widget.ConstraintLayout> </com.google.android.material.card.MaterialCardView> </pre>
--	--

Tabel 10 Source Code Jawaban Soal 1

B. Output Program



Gambar 2 Screenshot Hasil Jawaban Soal 1 XML



Gambar 3 Screenshot Hasil Jawaban Soal 1 Jetpack Compose

C. Pembahasan

GamesViewModel merupakan ViewModel yang menyimpan dan mengelola data game Roblox. ‘_games’ merupakan StateFlow privat yang menyimpan list game Roblox.

Lalu ‘games’ itu StateFlow publik yang bisa di-observe oleh UI, berisi daftar game. Kemudian ‘_selectedGame’ berfungsi sebagai StateFlow privat untuk menyimpan game yang dipilih user. ‘selectedGame’ juga StateFlow tapi publik yang bisa di-observe oleh UI, berisi game yang sedang dipilih. Terakhir ada fungsi ‘getGameById’ yang berguna untuk mencari game berdasarkan id. Mengembalikan objek RobloxGame yang sesuai, atau null jika tidak ditemukan.

Lalu pada NavGraph.kt, Sealed class digunakan untuk mendefinisikan semua rute yang ada di aplikasi. Setiap objek di dalamnya mewakili satu halaman (screen) dengan properti route. object GamesList disini mewakili halaman daftar game, dengan rute "games_list".

Object NavArguments merupakan objek untuk mendefinisikan nama argumen yang digunakan dalam navigasi, agar konsisten dan tidak typo. Terakhir ‘GAME_ID’ yang merupakan konstanta string "gameId" yang digunakan sebagai nama parameter pada rute detail game.

Pada RobloxGame.kt, ini merupakan kode yg berfungsi sebagai sumber data dummy di dalam aplikasi. Pertama, ia mendefinisikan sebuah data class bernama RobloxGame yang berfungsi sebagai "cetakan" atau template untuk menyimpan detail informasi sebuah game. Kemudian, ia membuat sebuah object bernama RobloxGamesData yang bertindak sebagai wadah terpusat. Di dalam wadah ini, dibuatlah sebuah daftar statis bernama games yang diisi dengan beberapa objek RobloxGame konkret.

Pada GameDetailScreen.kt, secara fungsional kode membangun sebuah layar detail untuk game yang bekerja secara dinamis dengan menerima sebuah gameId sebagai input. Layar tersebut kemudian berinteraksi dengan GamesViewModel untuk mengambil data lengkap game yang sesuai. Struktur visualnya diatur menggunakan Scaffold yang memiliki TopAppBar berisi judul game dan tombol navigasi untuk kembali ke layar sebelumnya. Seluruh konten detail seperti gambar header dengan efek gradien, kartu informasi berisi genre dan rating, serta deskripsi lengkap tersusun rapi secara vertikal dalam sebuah Column yang bisa digulir. Sebagai fitur utama, sebuah tombol "Play Now" akan menjalankan sebuah Intent sistem Android untuk membuka URL game di aplikasi lain. Kode ini juga menerapkan praktik yang baik dengan memecah tampilan menjadi komponen-komponen kecil yang dapat digunakan kembali seperti InfoColumn dan VerticalDivider demi menjaga kebersihan dan efisiensi kode.

Kemudian pada GameListScreen.kt, kode ini membangun sebuah layar utama yang menampilkan daftar game secara efisien menggunakan komponen LazyColumn, yang hanya merender item yang terlihat di layar sehingga sangat optimal untuk daftar panjang. Layar ini mengambil daftar data game dari sebuah GamesViewModel dan untuk setiap data game tersebut, ia membuat sebuah komponen kartu bernama GameCard. Komponen GameCard ini dirancang untuk menampilkan informasi ringkas satu game seperti gambar, judul, dan genre, serta menyediakan dua tombol interaktif. Fungsi dari kedua tombol ini didefinisikan di layar utama; tombol 'Play Now' akan memicu sebuah Intent untuk membuka URL game di aplikasi

eksternal, sedangkan tombol 'View Details' akan memerintahkan NavController untuk berpindah ke halaman detail sambil membawa ID game yang dipilih.

D. Tautan Git

Berikut adalah tautan untuk source code yang telah dibuat.

<https://github.com/NaufalElyzar11/PraktikumMobile/tree/main/Modul%204>

SOAL 2

Jelaskan Application class dalam arsitektur aplikasi Android dan fungsinya

A. Pembahasan

Application Class merupakan titik tertinggi atau Pusat dari aplikasi Android. Ini adalah komponen pertama yang dibuat saat aplikasi dimulai, bahkan sebelum Activity, Service, atau komponen lainnya. Karena ia dibuat pertama kali dan dihancurkan paling akhir, ia berfungsi sebagai tempat untuk menyimpan state dan logika yang bersifat global dan berlaku untuk keseluruhan siklus hidup aplikasi.